

**TED(10) - 4068
(REVISION-2010)**

**MODEL QUESTION PAPER (REV-10)
SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY
(Common for BME AND MD)**

MEDICAL IMAGING TECHNIQUES

(Maximum Marks : 100)

TIME : 3HRS

PART - A

Answer all Questions in one or two sentences. Each question carries **2 marks**.

I.

- a) List the different types of X-ray tubes used in radiography?
- b) Define Doppler ultrasound?
- c) Define FID signal?
- d) Name any four radioisotopes used in nuclear imaging?
- e) State the principle of generation of ultrasound?

5x2=10 marks

PART - B

Answer any **FIVE** of the following questions. Each question carries **6 marks**.

II.

- a) What are X rays? List the properties of X rays?
- b) Explain the procedures involved in balloon angioplasty?
- c) Explain echoencephalography?
- d) Comment on the applications of ultrasound in medical field?
- e) Explain how FID signal is obtained in NMR?
- f) List the properties of gamma rays?
- g) Account on the principle of PET Scanning?

5x6=30 marks

PART - C

Answer one full question from each Unit. Each question carries 15 marks.

UNIT I

- III** a) Explain the working of stationary anode X ray tube? **8**
- b) Explain the Computed Tomography? **7**

OR

- IV.** a) Explain the X ray machine with a neat block diagram? **10**
- b) What are stents? Mention its uses? **5**

UNIT II

- V.** a) Write the properties of ultrasound and how it is generated? **8**
- b) Give an account on A mode scan? **7**

OR

- VI** a). Explainechocardiography with a neat block diagram? **10**
- b) Mention the different types of scans used in ultrasound? **5**

UNIT III

VII a) Explain the principle of NMR imaging? **15**

OR

VIII a) Draw a neat block diagram of NMR system and explain the system components? **15**

UNIT IV

IX a) State the concept of nuclear imaging? **8**

b) Explain Anger Logic used in gamma camera? **7**

OR

X a) Explain the principle and procedure of SPECT scanning? **10**

b) List the applications of nuclear imaging? **5**

15x4=60